

Lecture 1

Automation and PLC

- What can be automated?
- Automation Technologies and Concepts
- PLC and PLC-Systems
 - Signal interfaces I/O
 - Sensors
 - Actuators
- PLC is a time discrete real-time system
- IEC 61131-3 and programming language

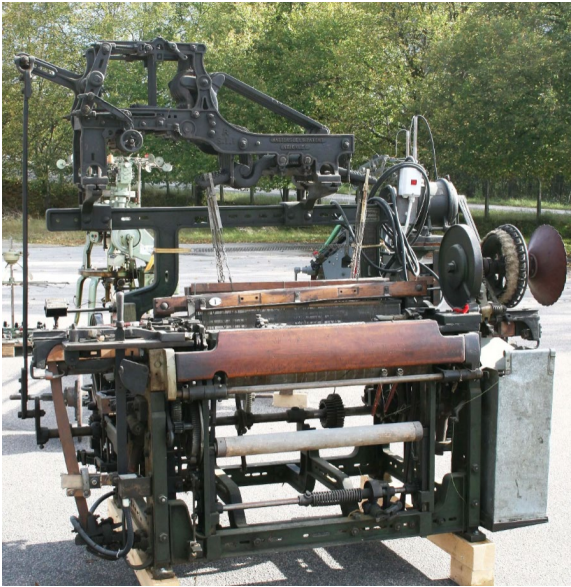
What can be automated?



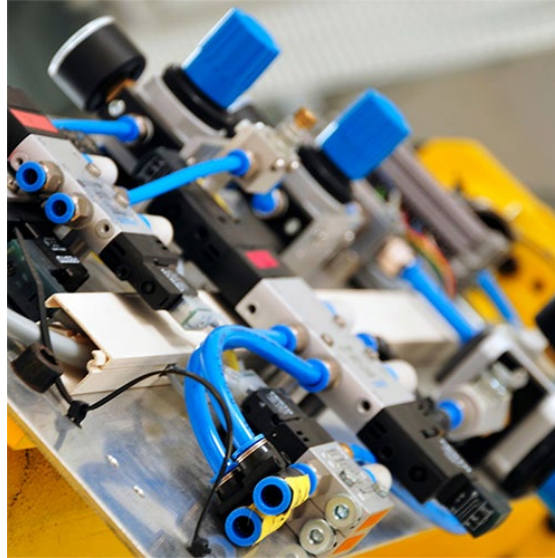
- In society
 - Healthcare
 - At home
 - Traffic
 - Road, Railroad, Sea, and Air

- In industry
 - Food
 - Medicine
 - Chemical
 - Manufacturing
 - Textile
 - Agriculture

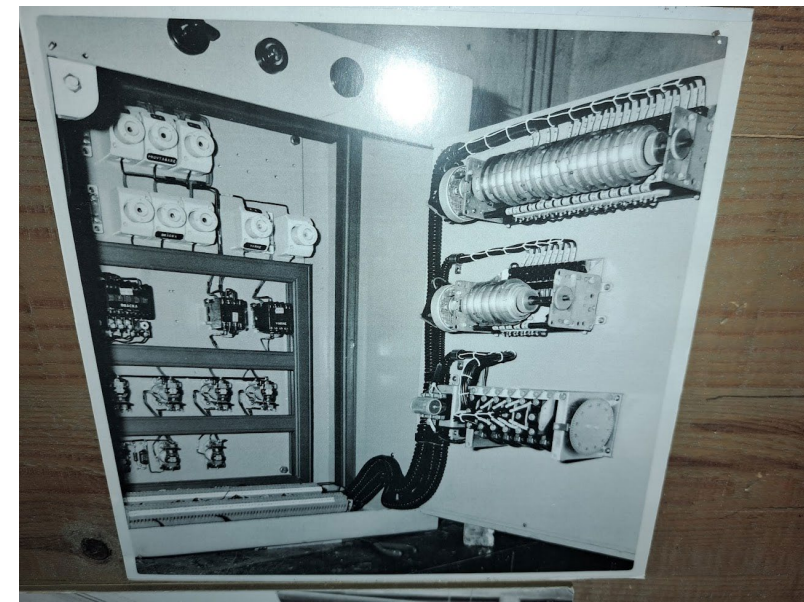
Automation techniques



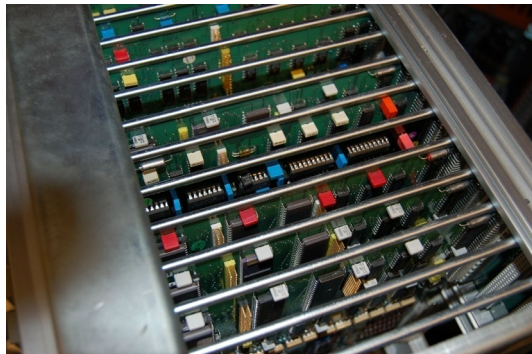
Mechanical



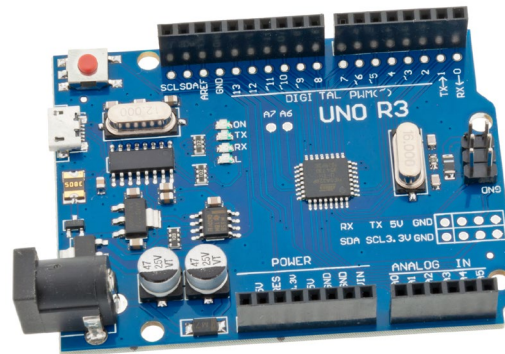
Pneumatical



Electromechanical



Electronical
Fixed logic

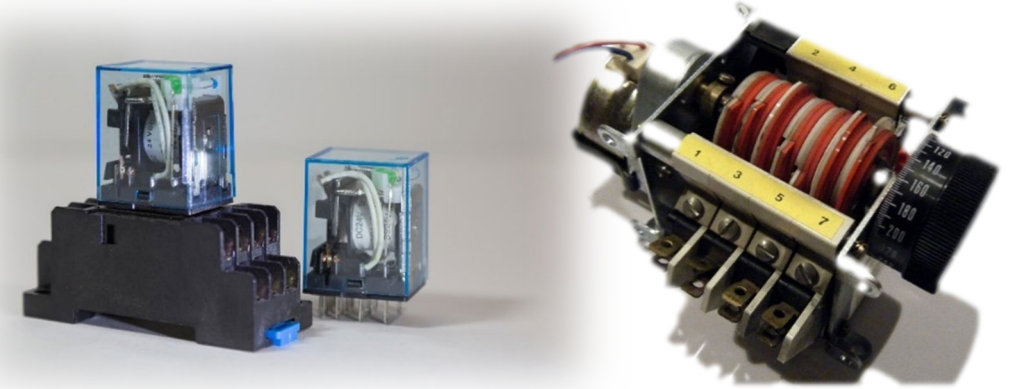


Micro controllers
Arduino, Raspberry PI, PC



Programmable Logic Control (PLC)

The origin of PLC



By Fylyp22 - Own work, CC BY 3.0,
<https://commons.wikimedia.org/w/index.php?curid=12180099>

- Updating of relay-based systems was time-consuming and expensive
 - Change function by rewiring each relay individually
 - Rebuild or replace timers and sequencers
- First PLC developed for the automobile industry (GM, US, 1968)
 - Modicon 084
- Replace hard-wired relays, cam timers, and drum sequencers

Programmable Logic Controller (PLC)



- An industrial computer
 - ruggedized and adapted for control of manufacturing processes
- Designed for application that requires
 - high reliability control
 - long-time stability
 - ease of programming
 - process fault diagnosis
- “Hard” real-time system
 - output result as response to input change within a specified time

Important Words

- | | |
|--------------------------|--|
| • Manual control | Requires Operator |
| • Automatic control | Manages itself (requires human monitoring) |
| • Open loop | No Sensors |
| • Closed Loop | Feedback through Sensors |
| • Discrete Logic Control | Digital (True / False), Boolean algebra |
| • Continuous Control | Binary coded or Analog control of entities |

Continuous and Discrete control



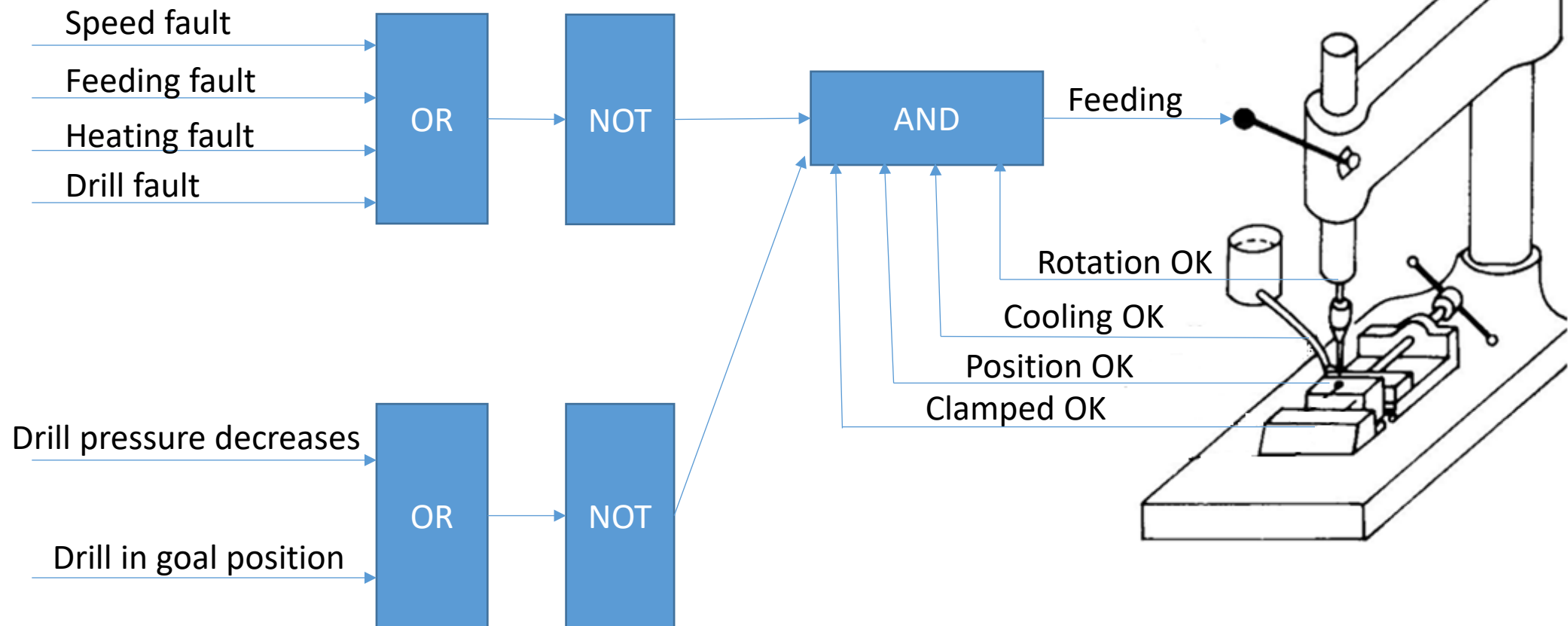
- PID - Controller
 - Continuous process control

- PLC
 - Control of discrete manufacturing



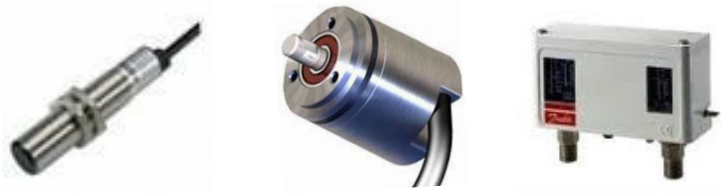
- Modern PLC
 - Handle both discrete and continuous processes
 - Store data internally or connected to a database
 - Communicate with other systems
 - Interacting with operators by Operator Panels (HMI)

Boolean Algebra (Discrete control)

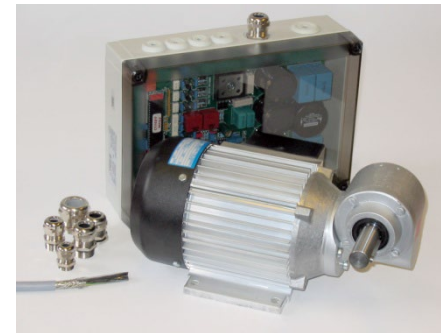


Actuators engaged by out-signals from the PLC

Sensors send in-signals to the PLC



Inductive sensor
Angle sensors
Pressure sensor



Motor control Frequency inverter
Valve
Relay/Contactor

Check this video in pneumatic or hydraulic controlled actuators
<https://www.youtube.com/watch?v=bXXL-0sf8gs>

Standards of signals

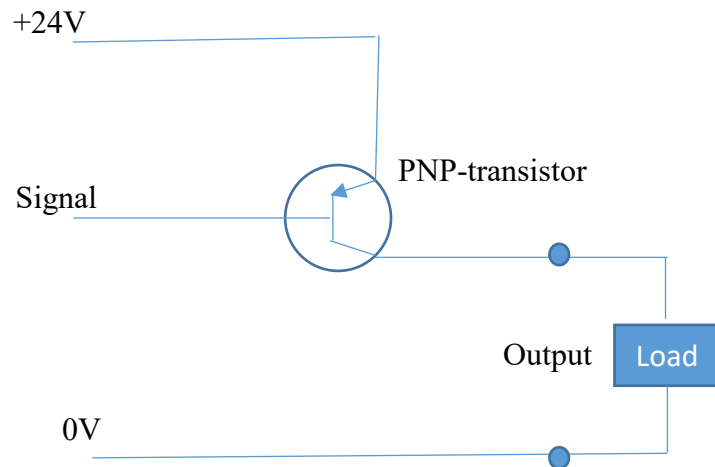
- Digital Signals
 - 24VDC Positive logic Industry standard in Europe
 - 24VDC Negative logic Common in Japan Asia
 - 110VAC Positive logic Old industry standard in the United States
 - 5VDC Logic & Computer Context Often negative logic
- Continuous Analog Signals
 - 0 to 20mA
 - 4 to 20mA
 - -10 to + 10V

Denomination of Digital and Analog I/O

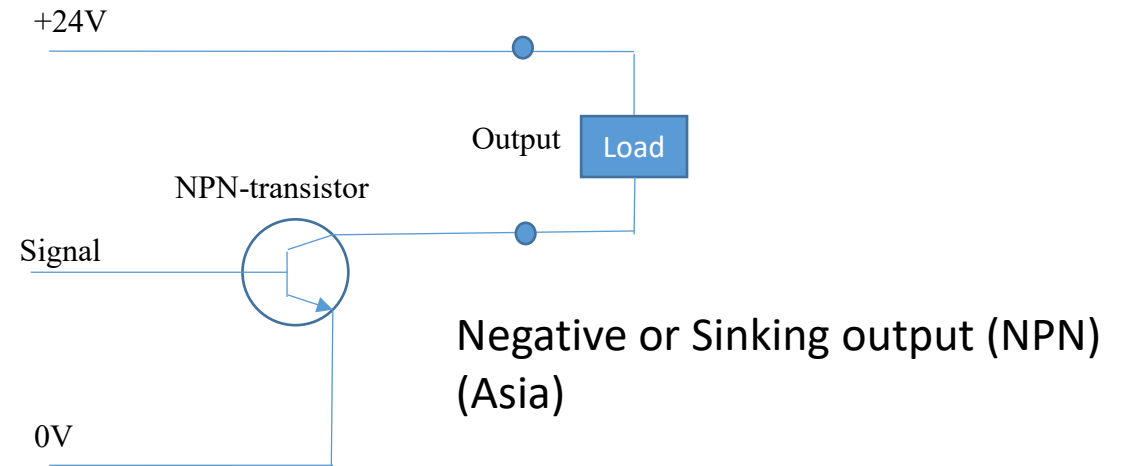
- Digital Inputs (IX)
- Digital Outputs (QX)
- Analog Inputs converted to Bytes, Word (IB, IW)
- Analog Output converted from Bytes, Word (QB, QW)
 - X = Bit 1-bit
 - B = Byte 8-bits
 - W = Word 16-bits
- Inputs and Outputs are traditionally numbered Octal or Hexadecimal starting with the number Zero

Two variants of Outputs

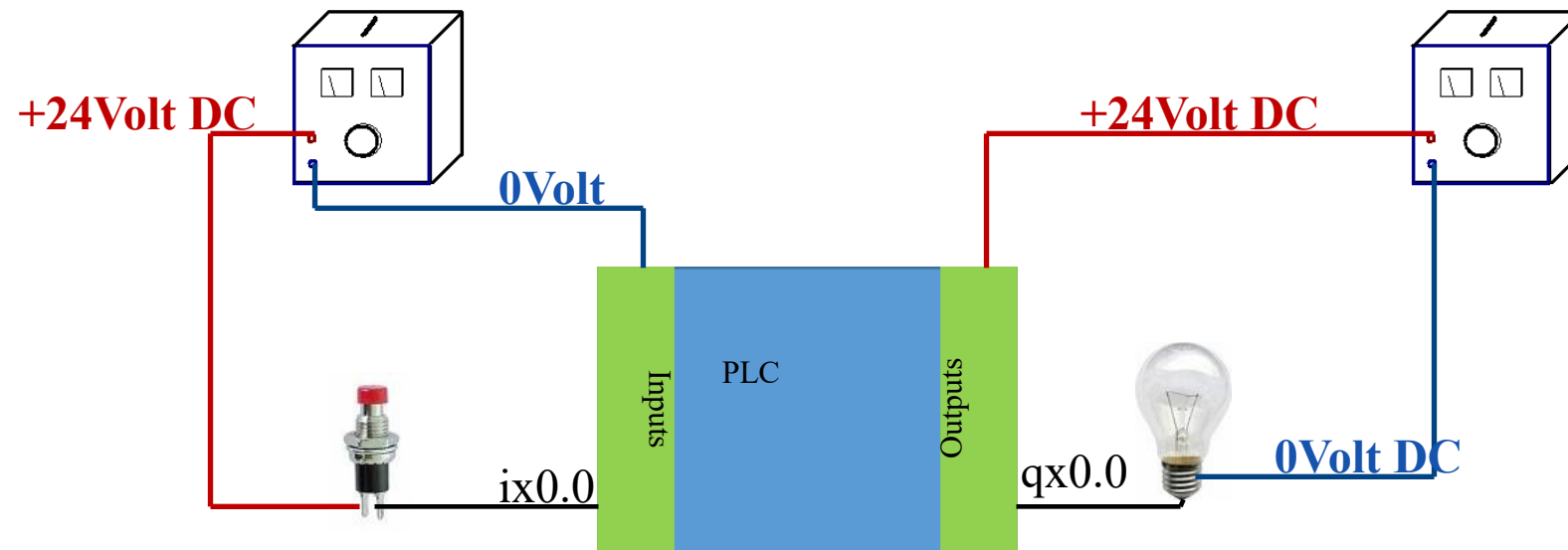
(From Sensors or PLC)



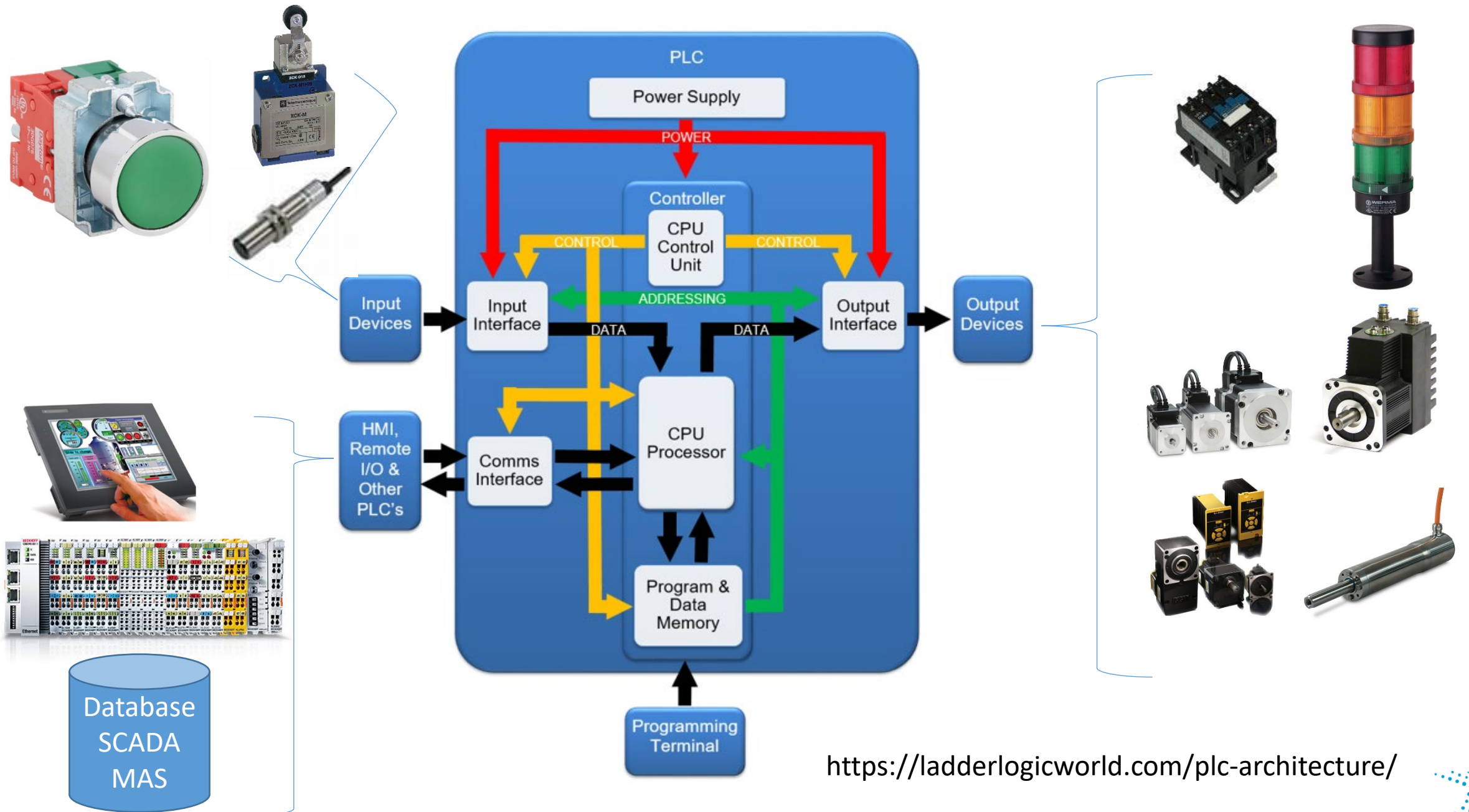
Positive or Sourcing output (PNP)
(Europe USA)



Connection of a PLC according to European Standard 24 Volt Direct Current (VDC) and Positive Logic (PNP)

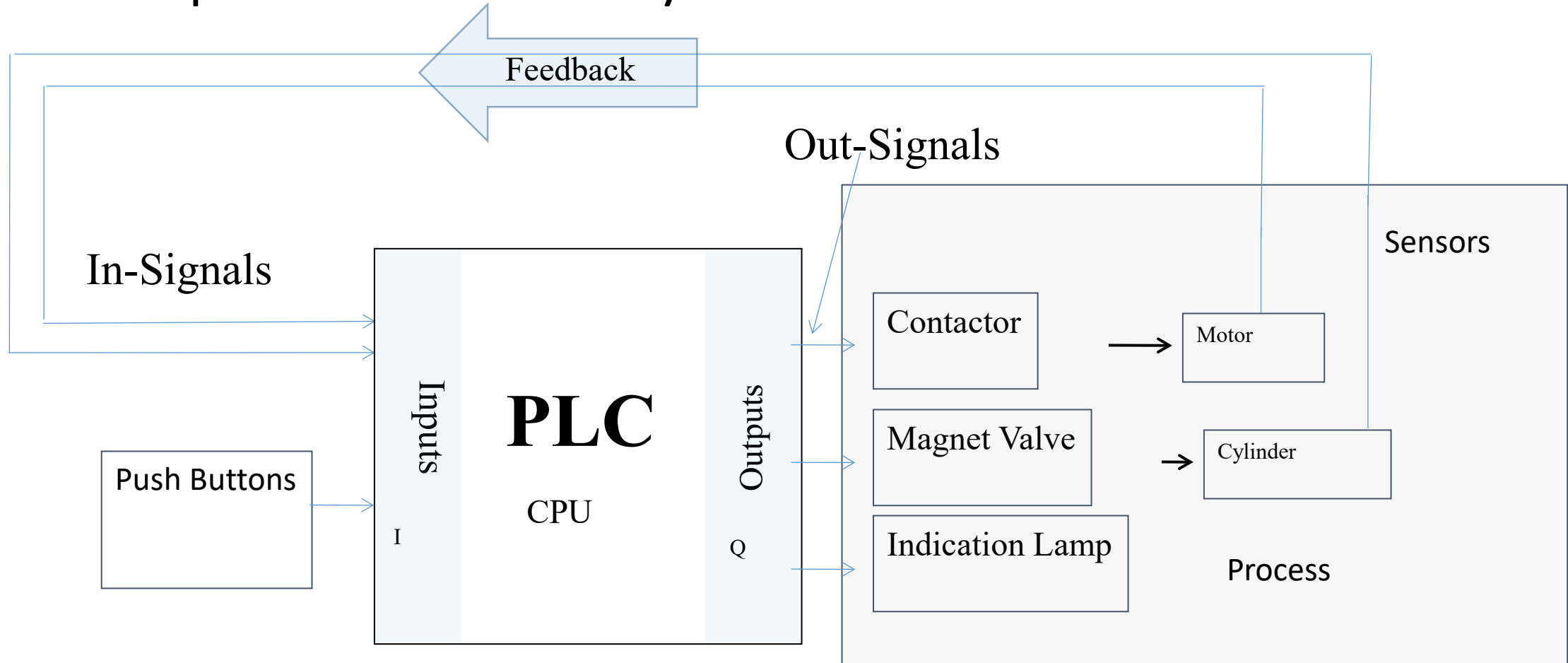


24 Volt is safe, you can't feel it!



<https://ladderlogicworld.com/plc-architecture/>

Principle of a PLC- System



Logic module (Simple PLC)



- Fits in a fuse box (230VAC)
- Boolean Logic only and some Fixed Functions as
 - Timers
 - Weekly calendars
- Perfect for controlling of
 - Pellet boiler
 - Staircase light
 - Outlet for car engine heater
- Very low price

Compact PLC



- Full-fledged PLC in a compact unit
- Can perform Boolean and Numerical Algebra
- Fixed number of I / O
 - e.g., 16 inputs and 8 outputs and often some analog signals
- Relatively low price

Rack based PLC (History)



- Usually one but sometimes several CPUs (multiprocessor systems)
- Different types of I / O boards available (digital usually 24VDC or analog e.g. 0-10V 4-20mA)
- Fixed maximum number of I/O cards
- Can often be expanded with more racks via fieldbus
- Is almost completely gone from the market today

Modular PLC

Distributed I/O



- Highly flexible system
- Can theoretically expand into infinity
- Large number of different I/O modules
- Several types of Fieldbus connections and other communications
- Easier electrical mounting than the rack-based systems

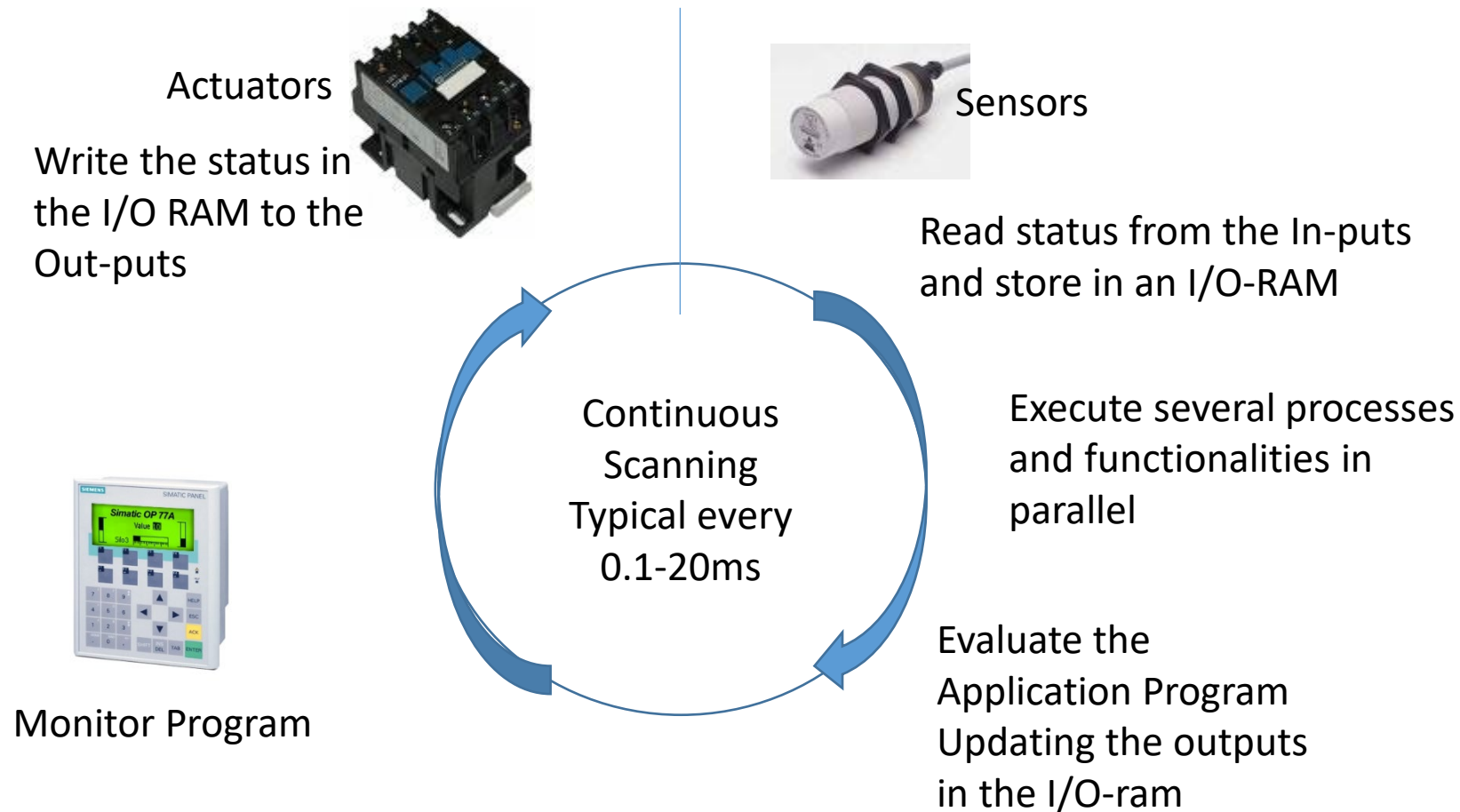
Soft-PLC



- A PC or Industrial PC (IPC) that executes the PLC program
- Windows or Linux
- Distributed I/O modules are connected via Ethernet
- Separate Real-Time core makes the system fast



PLC a time discrete system



PLC, a Real-Time system

- Real-time means known response time
- A PLC scans the program at regular intervals (Time Discrete System)
- The scan time in a PLC depends on the CPU speed and the number of program instructions executed during each scan
- A PLC can have:
 - Fast counter inputs count fast pulses separately and send the counter value to the PLC

IEC 61131 - 3

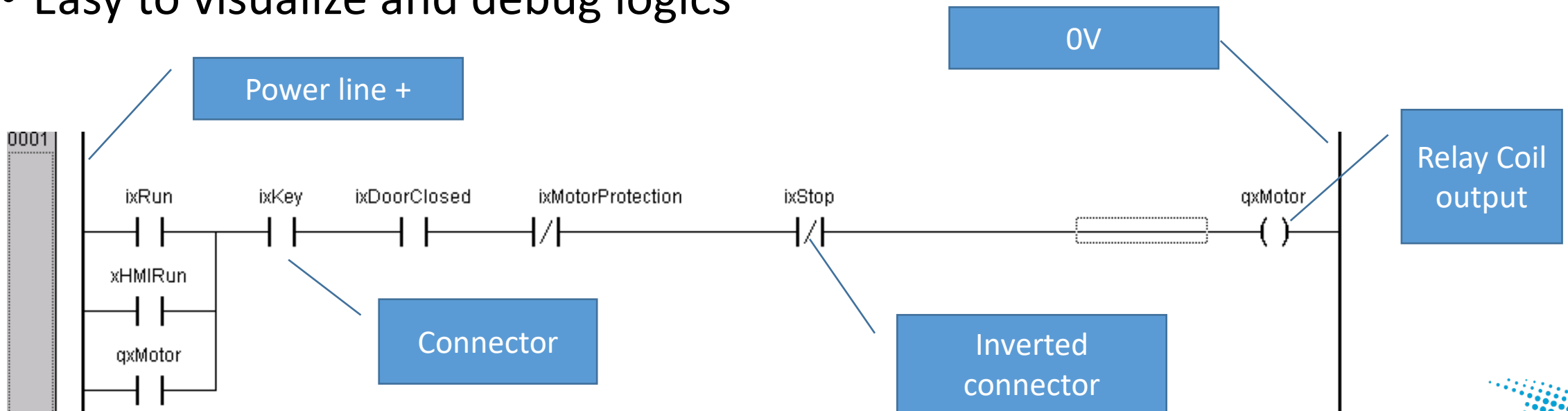
- A standard for “PLC programming”
- An open known and accepted standard
- Used by many suppliers
- Brand and hardware independent
- Cross platform reusability of logics
- Based on well tested methods and techniques
- Deterministic behavior

IEC 61131-3 languages

- Not a language standard –
a standard that includes more than only a language
- Five (Six) different languages included
 - each language has its purpose
 - hard to replace by one “super language”
 - historical reasons
 - some possibilities to alter between language (LD,FBD,IL)

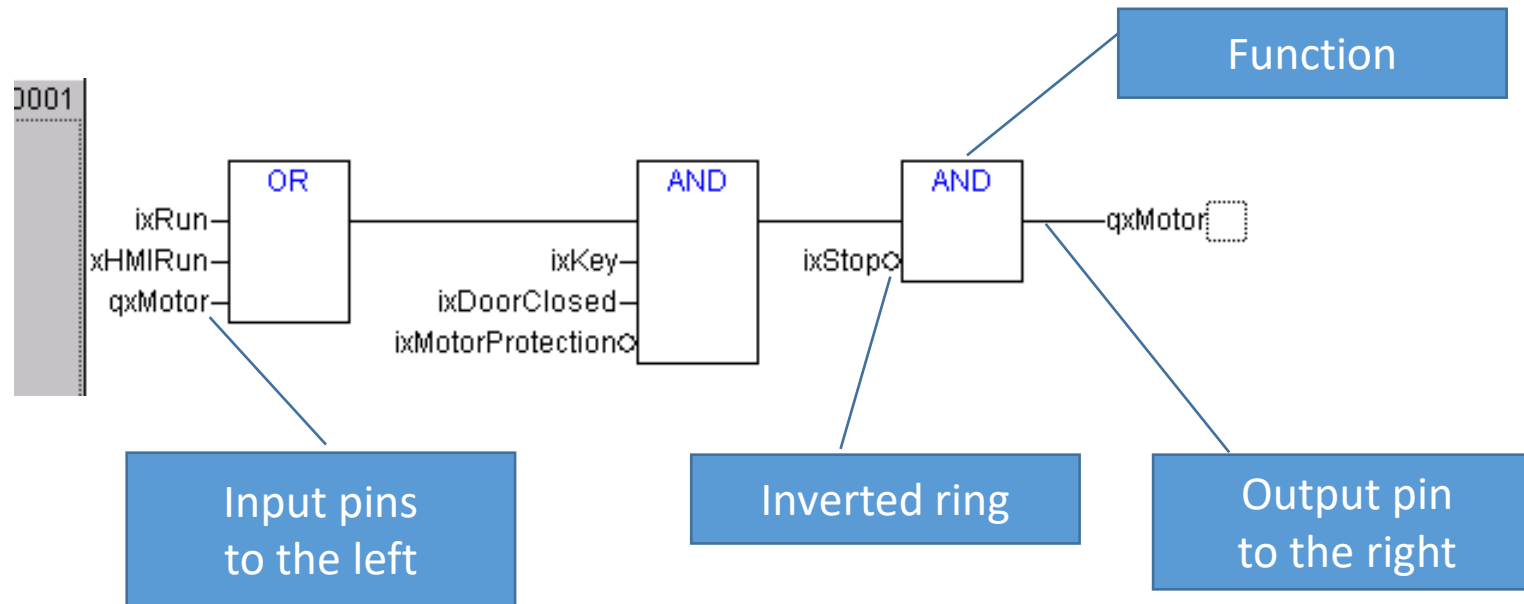
LD – Ladder Diagram

- Graphical symbols (relay chart)
- Boolean logic expressions
- Executed rung by rung (network by network)
- Easy to visualize and debug logics



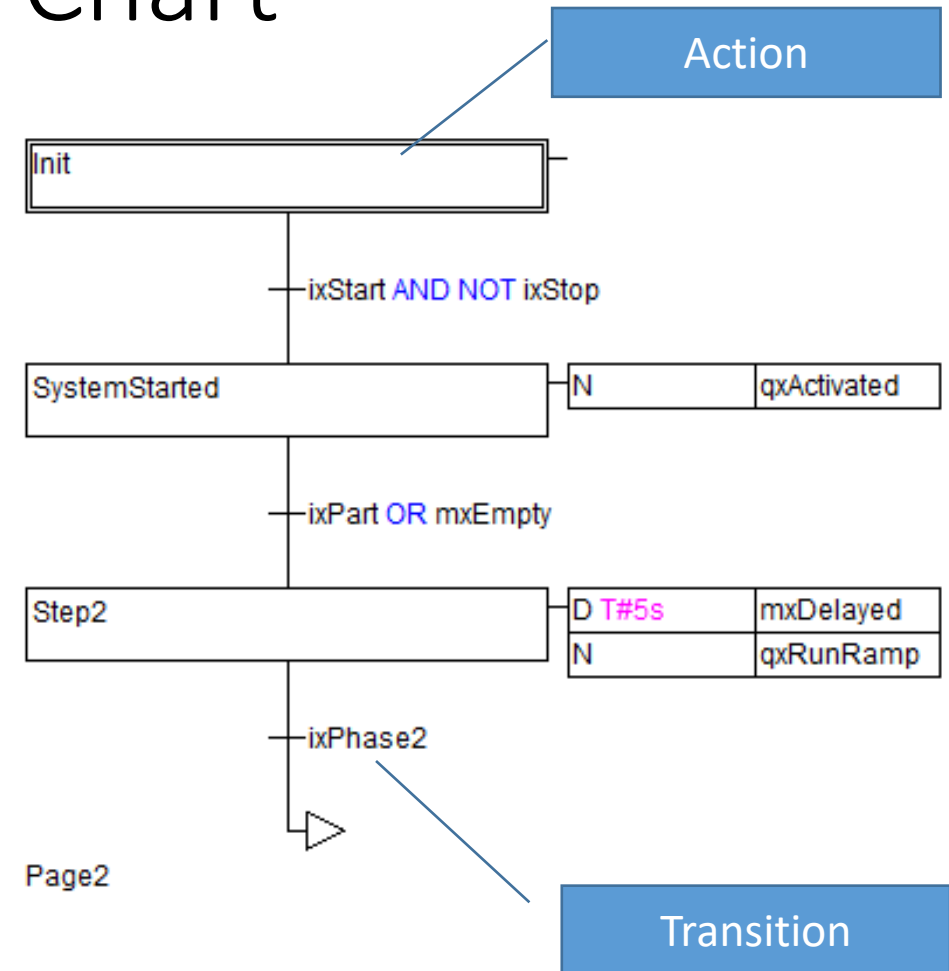
FBD – Function Block Diagram

- Graphical symbols (electronic chart)
- Same idea and design as LD
- Executed rung by rung



SFC – Sequential Function Chart

- Graphically oriented language
- Sequence of processing controlled by transition condition
- Action(s) with qualifier assigned to step elements



ST – Structured Text

- High-level “normal computer” programming language
 - Similar to Pascal, ABB-Rapid, and C ...
 - Assigns variable from right to left $a := b$;
 - The value in b will be assigned to a
 - Compare variables $a = b$
 - Is TRUE if $(a = b)$

```
qxMotor := ( ixStart OR qxMotor) AND NOT ixStop AND mxSafetyOk;
```

```
IF ixFilling AND ixLevel3 THEN  
    miLevel := 3;  
END_IF;
```

Read Chapter 1 and Chapter 2 In the book Programmable Logic Controller

Dag H. Hansen

Chapter 1

- PLC
 - History
 - Structure
 - Inputs
 - Outputs
 - Operations
 - **Do the Test Problems**

Chapter 2

- Digital Signals and Digital Inputs
 - Sensors
 - Switches
 - Actuators